

ME40 Thermodynamics Formula Sheet

General Relationships

- Temperature conversion: $T(K) = T(^{\circ}C) + 273.15$
 - Extensive property: Z
 - Intensive property: $z = \frac{Z}{m}$
 - State postulate: $z = z(x, y)$
 - Definition of specific enthalpy: $h = u + pv$
 - Volumetric conversions: $1 \text{ L} = 1000 \text{ dm}^3 = 0.001 \text{ m}^3$
 - Interpolation: $y = y_0 + (x - x_0) \frac{y_1 - y_0}{x_1 - x_0}$
 - Density and specific volume: $v = \frac{1}{\rho}$
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Ideal Liquids

- Incompressible fluid: $v = \text{const.}$
 - Heat capacity of liquid: $c_p(T) = c_v(T) = c(T)$
 - Specific internal energy change: $u(T_2) - u(T_1) = \int_{T_1}^{T_2} c(T) dT$
 - Specific enthalpy change: $h(T_2) - h(T_1) = \int_{T_1}^{T_2} c(T) dT + v(P_2 - P_1)$
 - Specific entropy change: $s_2(T_2) - s_1(T_1) = \int_{T_1}^{T_2} \frac{c(T)}{T} dT$
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Ideal Gases

- Ideal gas constant: $\bar{R} = 8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}},$
 $R = \frac{\bar{R}}{M} \text{ (} M = \text{molar mass)}$
- Ideal gas law: $Pv = RT \quad (\text{Molar Version: } PV = n\bar{R}T)$
- Specific heat capacity at constant volume: $c_v = \left(\frac{\partial u}{\partial T} \right)_v$
- Specific heat capacity at constant pressure: $c_p = \left(\frac{\partial h}{\partial T} \right)_p$
- Specific heat ratio: $k = \frac{c_p}{c_v}$
- Specific heat relation: $R = c_p - c_v$
- Specific internal energy change: $u(T_2) - u(T_1) = \int_{T_1}^{T_2} c_v(T) dT$

- Specific enthalpy change: $h(T_2) - h(T_1) = \int_{T_1}^{T_2} c_p(T) dT$
 - Specific entropy change : $s(T_2, p_2) - s(T_1, p_1) = \int_{T_1}^{T_2} \frac{c_p(T)}{T} dT - R \ln \frac{p_2}{p_1}$
 - Specific entropy change (Tables): $s(T_2, p_2) - s(T_1, p_1) = s^0(T_2) - s^0(T_1) - R \ln \frac{p_2}{p_1}$
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Real Fluids

- Property tables: $z = f(T, p) \quad (z = v, u, h, s)$
 - Saturated mixture properties: $z = z_f + x(z_g - z_f)$
 - Saturated mixture quality: $x = \frac{m_{\text{gas}}}{m_{\text{liquid}} + m_{\text{gas}}}$
 - Heat of vaporization: $h_{fg} = h_g - h_f$
 - Specific volume change of vaporization: $v_{fg} = v_g - v_f$
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Processes

- Isobaric: $p = \text{const.}$
 - Isothermal: $T = \text{const.}$
 - Isochoric: $V = \text{const.}$
 - Isenthalpic: $H = \text{const.}$
 - Isentropic: $S = \text{const.}$
 - Polytropic: $pV^n = \text{const.}$ (n - polytropic coefficient)
 - Isentropic ideal gas (constant specific heats): $\left(\frac{T_2}{T_1}\right) = \left(\frac{v_1}{v_2}\right)^{k-1} = \left(\frac{P_2}{P_1}\right)^{(k-1)/k} ; k = \frac{c_p}{c_v}$
 - Isentropic ideal gas (variable specific heats): $s_2 = s_1 + R \ln \frac{P_2}{P_1} ; \left(\frac{P_2}{P_1}\right) = \frac{P_{r2}}{P_{r1}} ; \left(\frac{v_2}{v_1}\right) = \frac{v_{r2}}{v_{r1}}$
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Energy

- Total energy: $E = U + KE + PE$
- Change in total energy: $\Delta E = \Delta U + \Delta KE + \Delta PE$
- Specific kinetic and potential energies: $ke = \frac{w^2}{2}, \quad pe = gz$

Boundary Work

- General boundary work: $W_{12} = \int_1^2 P dV$
- Isothermal boundary work: $W_{12} = P_1 V_1 \ln \frac{V_2}{V_1} = P_1 V_1 \ln \frac{P_1}{P_2}$
- Polytropic boundary work: $W_{12} = \frac{1}{1-n} (P_2 V_2 - P_1 V_1)$
- Ideal gas polytropic boundary work: $W_{12} = \frac{R(T_2 - T_1)}{1-n}$

Reversible Flow Work

- Reversible steady-flow work: $w_{\text{rev}} = - \int_1^2 v dP - \Delta \text{ke} - \Delta \text{pe}$
- Reversible steady-flow work (incompressible): $w_{\text{rev}} = -v(P_2 - P_1) - \Delta \text{ke} - \Delta \text{pe}$

Energy and Energy Balances

- Potential energy of a spring: $E = \frac{k(x-x_0)^2}{2}$ with $F = k(x - x_0)$
- Energy of a closed system: $\Delta E = Q - W, \quad \frac{dE}{dt} = \dot{Q} - \dot{W}$
- Energy balance for cycles: $W_{\text{Cycle}} = Q_{\text{in}} - Q_{\text{out}}$
- Energy of an open system:
$$\Delta E = Q - W + \sum_{i=1}^n \Delta m_{i,\text{in}} \left(h_{i,\text{in}} + \frac{w_{i,\text{in}}^2}{2} + gz_{j,\text{in}} \right) - \sum_{j=1}^k \Delta m_{j,\text{ex}} \left(h_{j,\text{in}} + \frac{w_{j,\text{ex}}^2}{2} + gz_{j,\text{ex}} \right)$$
- Energy of an open system (rate):
$$\frac{dE}{dt} = \dot{Q} - \dot{W} + \sum_{i=1}^n \dot{m}_{i,\text{in}} \left(h_{i,\text{in}} + \frac{w_{i,\text{in}}^2}{2} + gz_{j,\text{in}} \right) - \sum_{j=1}^k \dot{m}_{j,\text{ex}} \left(h_{j,\text{in}} + \frac{w_{j,\text{ex}}^2}{2} + gz_{j,\text{ex}} \right)$$

Efficiency

- Thermal efficiency: $\eta_{\text{th}} = \frac{W_{\text{net, out}}}{Q_{\text{H}}} = 1 - \frac{Q_{\text{L}}}{Q_{\text{H}}}$
- Coefficient of performance (refrigerator): $\text{COP}_{\text{R}} = \frac{Q_{\text{L}}}{W_{\text{net, in}}} = \frac{1}{Q_{\text{H}}/Q_{\text{L}} - 1}$
- Coefficient of performance (heat pump): $\text{COP}_{\text{HP}} = \frac{Q_{\text{H}}}{W_{\text{net, in}}} = \frac{1}{1 - Q_{\text{L}}/Q_{\text{H}}}$
- Thermal efficiency (reversible): $\eta_{\text{th}} = 1 - \frac{T_{\text{L}}}{T_{\text{H}}}$
- Coefficient of performance (reversible refrigerator): $\text{COP}_{\text{R, rev}} = \frac{1}{T_{\text{H}}/T_{\text{L}} - 1}$
- Coefficient of performance (reversible heat pump): $\text{COP}_{\text{HP, rev}} = \frac{1}{1 - T_{\text{L}}/T_{\text{H}}}$

- Isentropic efficiency (turbine): $\eta_T = \frac{\text{Actual turbine work}}{\text{Isentropic turbine work}} = \frac{w_a}{w_s} \approx \frac{h_1 - h_{2a}}{h_1 - h_{2s}}$
 - Isentropic efficiency (compressor): $\eta_C = \frac{\text{Isentropic compressor work}}{\text{Actual compressor work}} = \frac{w_s}{w_a} \approx \frac{h_{2s} - h_1}{h_{2a} - h_1}$
 - Isentropic efficiency (nozzle): $\eta_T = \frac{\text{Actual KE at nozzle exit}}{\text{Isentropic KE at nozzle exit}} = \frac{V_{2a}^2}{V_{2s}^2} \approx \frac{h_1 - h_{2a}}{h_1 - h_{2s}}$
 - Second law efficiency: $\eta_{II} = \frac{\text{Exergy recovered}}{\text{Exergy expended}} = 1 - \frac{\text{Exergy destroyed}}{\text{Exergy expended}}$
 - Second-law efficiency (work-producing): $\eta_{II} = \frac{\eta_{th}}{\eta_{th,rev}} = \frac{W_u}{W_{rev}}$
 - Second-law efficiency (work-consuming): $\eta_{II} = \frac{COP}{COP_{rev}} = \frac{W_{rev}}{W_u}$
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Entropy

- Entropy balance closed systems: $S_{gen} = S_2 - S_1 - \int_1^2 \frac{\delta Q}{T_B}$
 - Entropy balance fixed boundary Temperature T_B closed systems: $S_{gen} = S_2 - S_1 - \sum_{j=1}^n \frac{Q_j}{T_{B,j}}$
 - Unsteady process: $S_{gen} = S_2 - S_1 - \sum_{j=1}^n \frac{Q_j}{T_{B,j}} + \sum_{k=1}^m \Delta m_{e,k} s_{e,k} - \sum_{i=1}^l \Delta m_{i,i} s_{i,i}$
 - Entropy balance control volumes: $\dot{S}_{gen} = \frac{dS}{dt} + \sum_{k=1}^m \dot{m}_{e,k} s_{e,k} - \sum_{i=1}^l \dot{m}_{i,i} s_{i,i} - \sum_{j=1}^n \frac{Q_j}{T_{B,j}}$
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Exergy

- Exergy of a stationary fluid: $\phi = (u - u_0) + P_0(v - v_0) - T_0(s - s_0) + \frac{V^2}{2} + gz$
- Exergy of a flowing fluid: $\psi = (h - h_0) - T_0(s - s_0) + \frac{V^2}{2} + gz$
- Change in exergy: $\Delta X = X_2 - X_1 = m(\phi_2 - \phi_1)$
 $\Delta \psi = \psi_2 - \psi_1$
- Exergy transfer by heat: $X_{heat} = \int_1^2 \left(1 - \frac{T_0}{T}\right) \delta Q$
 $\dot{X}_{heat} = \left(1 - \frac{T_0}{T}\right) \dot{Q}$
- Exergy destruction: $X_{destroyed} = T_0 S_{gen} = W_{rev,out} - W_{u,out} = W_{u,in} - W_{rev,in}$
- Exergy balance: $X_2 - X_1 = \int_1^2 \left(1 - \frac{T_0}{T}\right) \delta Q - [W - P_0(V_2 - V_1)] + \sum_{in} m\psi - \sum_{out} m\psi - X_{destroyed}$
- Exergy balance rate formulation: $\frac{dX_{CV}}{dt} = \sum \left(1 - \frac{T_0}{T_k}\right) \dot{Q}_k - \left(\dot{W} - P_0 \frac{dV_{CV}}{dt}\right) + \sum_{in} m\psi - \sum_{out} m\psi = \dot{X}_{destroyed}$

Thermodynamic Property Relations

- Fundamental thermodynamic relations:

$$du = Tds - Pdv$$

$$dh = Tds + vdp$$

$$da = -sdT - Pdv$$

$$dg = -sdT + vdp$$

- Maxwell relations:

$$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial S}\right)_V$$

$$\left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P$$

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V$$

$$\left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P$$

Gas Mixtures

- Mass Fraction and Mole Fraction:

$$mf_i = \frac{m_i}{m_m} \text{ and } y_i = \frac{N_i}{N_m}$$

- Total Mass and Moles:

$$m_m = \sum_{i=1}^n m_i \text{ and } y_i = \sum_{i=1}^n N_i$$

- Molar Mass Equations:

$$M_m = \frac{m_m}{N_m} = \sum_{i=1}^n y_i M_i \text{ and } R_m = \frac{R_u}{M_m}$$

- Dalton's Law:

$$P_m = \sum_{i=1}^n P_i(T_m, V_m)$$

- Amagat's Law:

$$V_m = \sum_{i=1}^n V_i(T_m, P_m)$$

- Component Fractions for Ideal Gases:

$$\frac{P_i}{P_m} = \frac{V_i}{V_m} = \frac{N_i}{N_m} = y_i$$

- Extensive Properties(Z) of a Gas Mixture(U, H, S):

$$Z_m = \sum_{i=1}^k Z_i = \sum_{i=1}^k m_i z_i = \sum_{i=1}^k m_i \bar{z}_i$$

- Intensive Properties(z) of a Gas Mixture(u, h, s, c_v , c_p):

$$z_m = \sum_{i=1}^k mf_i z_i = \sum_{i=1}^k y_i \bar{z}_i$$

Gas-Vapor Mixtures and Air-Conditioning

- Enthalpy of Dry Air: $\Delta h_{\text{dry air}} = c_p \Delta T = 1.005 \Delta T$
- Atmospheric Air: $P = P_a + P_v$
- Enthalpy of Water vapor: $h_v(T, \text{low } P) \simeq 2500.8 + 1.82T$
- Specific or Absolute Humidity: $\omega = \frac{m_v}{m_a} = \frac{0.622 P_v}{P - P_v}$
- Relative Humidity: $\phi = \frac{m_v}{m_g} = \frac{P_v}{P_g}$
- Absolute and Relative Humidity Relation: $\phi = \frac{\omega P}{(0.622 + \omega)P}$ and $\omega = \frac{-0.622 \phi P}{P - \phi P_g}$
- Enthalpy of Atmospheric Air: $h = h_a + \omega h_g$

Chemical Reactions

- Air-Fuel Ratio: $\text{AF} = \frac{m_{\text{air}}}{m_{\text{fuel}}}$
- Conservation of Energy for an Open System: $Q - W = \sum N_p (\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_p - \sum N_r (\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_r$
- Conservation of Energy for a Closed System: $Q - W = \sum N_p (\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ - P\bar{V})_p - \sum N_r (\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ - P\bar{V})_r$
- Process Entropy: $\sum \frac{Q_k}{T_k} + S_{\text{gen}} = S_{\text{prod}} - S_{\text{react}}$
- Entropy for an Ideal Gas Mixture: $\bar{s}_i(T, P_i) = \bar{s}_i^\circ(T, P_0) - R_u \ln \frac{y_i P_m}{P_0}$
- Gibbs Function: $W_{\text{rev}} = \sum N_r (\bar{g}_f^\circ + \bar{g} - \bar{g}^\circ)_r - \sum N_p (\bar{g}_f^\circ + \bar{g} - \bar{g}^\circ)_p$